

2018 年数学(三)真题解析

一、选择题

(1) 【答案】 (D).

【解】 对选项 D, 由导数定义得

$$f'_+(0) = \lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{\cos \sqrt{x} - 1}{x} = \lim_{x \rightarrow 0^+} \frac{-\frac{1}{2}x}{x} = -\frac{1}{2},$$

$$f'_-(0) = \lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^-} \frac{\cos \sqrt{-x} - 1}{x} = \lim_{x \rightarrow 0^-} \frac{\frac{1}{2}x}{x} = \frac{1}{2},$$

故函数 $f(x) = \cos \sqrt{|x|}$ 不可导, 同理, 可验证其他 3 个选项在 $x = 0$ 都可导, 应选(D).

(2) 【答案】 (D).

【解】 考虑 $f(x)$ 在 $x = \frac{1}{2}$ 处的泰勒公式:

$$f(x) = f\left(\frac{1}{2}\right) + f'\left(\frac{1}{2}\right)\left(x - \frac{1}{2}\right) + \frac{f''(\xi)}{2!}\left(x - \frac{1}{2}\right)^2 \quad \left(\text{其中 } \xi \text{ 在 } x \text{ 与 } \frac{1}{2} \text{ 之间}\right).$$

对 $f(x)$ 在 $[0, 1]$ 上积分得

$$\begin{aligned} 0 &= \int_0^1 f(x) dx = f\left(\frac{1}{2}\right) + f'\left(\frac{1}{2}\right) \int_0^1 \left(x - \frac{1}{2}\right) dx + \frac{1}{2} \int_0^1 f''(\xi) \left(x - \frac{1}{2}\right)^2 dx \\ &= f\left(\frac{1}{2}\right) + \frac{1}{2} \int_0^1 f''(\xi) \left(x - \frac{1}{2}\right)^2 dx. \end{aligned}$$

所以, 当 $f''(x) > 0$ 时, $f\left(\frac{1}{2}\right) = -\frac{1}{2} \int_0^1 f''(\xi) \left(x - \frac{1}{2}\right)^2 dx < 0$, 应选(D).

(3) 【答案】 (C).

【解】 由定积分性质得 $M = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{(1+x)^2}{1+x^2} dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(1 + \frac{2x}{1+x^2}\right) dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} dx = \pi$,

又当 $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ 时, $1 + \sqrt{\cos x} > 1 > \frac{1+x}{e^x}$, 由定积分的保号性, 有

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1 + \sqrt{\cos x}) dx > \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 1 dx > \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1+x}{e^x} dx, \text{ 即 } K > M > N, \text{ 应选(C).}$$

(4) 【答案】 (D).

【解】 平均成本函数 $\bar{C}(Q) = \frac{C(Q)}{Q}$, 则 $\bar{C}'(Q) = \frac{C'(Q)Q - C(Q)}{Q^2}$.

由题设产量为 Q_0 时平均成本最小, 所以 $C'(Q_0) = \frac{C'(Q_0)Q_0 - C(Q_0)}{Q_0^2} = 0$,

即 $C'(Q_0)Q_0 = C(Q_0)$, 应选(D).

(5) 【答案】 (A).

【解】 记矩阵 $M = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$, 则 $|\lambda E - M| = \begin{vmatrix} \lambda - 1 & 1 & 0 \\ 0 & \lambda - 1 & 1 \\ 0 & 0 & \lambda - 1 \end{vmatrix} = (\lambda - 1)^3 = 0$.

所以矩阵 $\mathbf{M}$ 的特征值为 $\lambda_1 = \lambda_2 = \lambda_3 = 1$, 且 $r(\lambda \mathbf{E} - \mathbf{M}) = r(\mathbf{E} - \mathbf{M}) = 2$, 设选项 (A), (B), (C), (D) 的矩阵分别为 $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$, 易算出其特征值均为 1, 且 $r(\lambda \mathbf{E} - \mathbf{A}) = r(\mathbf{E} - \mathbf{A}) = 2$, $r(\mathbf{E} - \mathbf{B}) = r(\mathbf{E} - \mathbf{C}) = r(\mathbf{E} - \mathbf{D}) = 1$,

若两矩阵相似, 其对应的特征值矩阵也相似, 秩相等. 所以可以判断选项 (A) 正确, 应选 (A).

(6) 【答案】 (A).

【解】 易知 $r(\mathbf{A}, \mathbf{AB}) \geq r(\mathbf{A})$, 又由分块矩阵的乘法, 可知 $(\mathbf{A}, \mathbf{AB}) = \mathbf{A}(\mathbf{E}, \mathbf{B})$, 因此

$$r(\mathbf{A}, \mathbf{AB}) \leq \min\{r(\mathbf{A}), r(\mathbf{E}, \mathbf{B})\},$$

从而 $r(\mathbf{A}, \mathbf{AB}) \leq r(\mathbf{A})$, 所以 $r(\mathbf{A}, \mathbf{AB}) = r(\mathbf{A})$, 应选 (A).

(7) 【答案】 (A).

【解】 由 $f(1+x) = f(1-x)$, 知 $f(x)$ 关于 $x=1$ 对称, 因此

$$\int_0^1 f(x) dx = \int_1^2 f(x) dx = \frac{1}{2} \int_0^2 f(x) dx = 0.3,$$

$$P\{X < 0\} = \int_{-\infty}^0 f(x) dx = \int_{-\infty}^1 f(x) dx - \int_0^1 f(x) dx = 0.5 - 0.3 = 0.2.$$

故选 (A).

(8) 【答案】 (B).

【解】 根据单个正态总体的分布性质可知

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right), \quad \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1), \quad \frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1),$$

$$\text{且 } \bar{X} \text{ 与 } S^2 \text{ 独立, 所以 } \frac{\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}}{\sqrt{\frac{(n-1)S^2}{\sigma^2} \cdot \frac{1}{n-1}}} \sim t(n-1), \text{ 即 } \frac{\sqrt{n}(\bar{X} - \mu)}{S} \sim t(n-1),$$

应选 (B).

二、填空题

(9) 【答案】 $y = 4x - 3$.

【解】 $y' = 2x + \frac{2}{x}$, $y'' = 2 - \frac{2}{x^2}$ ($x > 0$). 由 $y'' = 2 - \frac{2}{x^2} = 0$ 得拐点为 $(1, 1)$.

又所求切线的斜率为 $y' = \left(2x + \frac{2}{x}\right) \Big|_{x=1} = 4$, 故切线方程为 $y = 4x - 3$.

(10) 【答案】 $e^x \arcsin \sqrt{1 - e^{2x}} - \sqrt{1 - e^{2x}} + C$.

【解】

$$\begin{aligned} \int e^x \arcsin \sqrt{1 - e^{2x}} dx &= \int \arcsin \sqrt{1 - e^{2x}} de^x \\ &= \arcsin \sqrt{1 - e^{2x}} \cdot e^x - \int \frac{1}{\sqrt{1 - (1 - e^{2x})}} \cdot \frac{1}{2\sqrt{1 - e^{2x}}} \cdot (-e^{2x} \cdot 2) \cdot e^x dx \\ &= e^x \arcsin \sqrt{1 - e^{2x}} + \int \frac{1}{\sqrt{1 - e^{2x}}} \cdot e^{2x} dx \\ &= e^x \arcsin \sqrt{1 - e^{2x}} - \sqrt{1 - e^{2x}} + C. \end{aligned}$$

(11) 【答案】 $C2^x - 5$ (C 为任意常数).

【解】 因为 $\Delta^2 y_x = y_{x+2} - 2y_{x+1} + y_x$, 故原方程化为 $y_{x+2} - 2y_{x+1} = 5$,

对应齐次方程的通解为 $\bar{y}_x = C \cdot 2^x$ (C 为任意常数).

设非齐次方程的特解为 $y_x^* = A$, 则 $A - 2A = 5$, 即 $A = -5$.

故所求通解为 $y_x = C2^x - 5$ (C 为任意常数).

(12) 【答案】 $2e$.

【解】 由 $f(x + \Delta x) - f(x) = 2xf(x)\Delta x + o(\Delta x)$ ($\Delta x \rightarrow 0$), 得

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = 2xf(x), \text{ 即 } f'(x) = 2xf(x).$$

解之得 $f(x) = Ce^{x^2}$ (C 为任意常数).

由于 $f(0) = 2$, 得 $C = 2$. 所以 $f(x) = 2e^{x^2}$, $f(1) = 2e$.

(13) 【答案】 2 .

【解】 由 $A\alpha_1 = \alpha_1 + \alpha_2, A\alpha_2 = \alpha_2 + \alpha_3, A\alpha_3 = \alpha_1 + \alpha_3$ 有

$$A(\alpha_1, \alpha_2, \alpha_3) = (A\alpha_1, A\alpha_2, A\alpha_3) = (\alpha_1, \alpha_2, \alpha_3) \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix},$$

两边取行列式, 得

$$|A| |\alpha_1, \alpha_2, \alpha_3| = |\alpha_1, \alpha_2, \alpha_3| \begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = |\alpha_1, \alpha_2, \alpha_3| \begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{vmatrix} = 2 |\alpha_1, \alpha_2, \alpha_3|.$$

已知 $\alpha_1, \alpha_2, \alpha_3$ 线性无关, 知 $|\alpha_1, \alpha_2, \alpha_3| \neq 0$, 所以 $|A| = 2$.

(14) 【答案】 $\frac{1}{3}$.

【解】

$$\begin{aligned} P(AC | A \cup B) &= \frac{P[AC(A \cup B)]}{P(A \cup B)} = \frac{P(AC \cup ABC)}{P(A \cup B)} = \frac{P(AC)}{P(A) + P(B) - P(AB)} \\ &= \frac{P(A)P(C)}{P(A) + P(B) - P(A)P(B)} = \frac{\frac{1}{2} \times \frac{1}{2}}{\frac{1}{2} + \frac{1}{2} - \frac{1}{2} \times \frac{1}{2}} = \frac{1}{3}. \end{aligned}$$

三、解答题

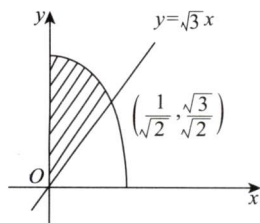
(15) 【解】 $\lim_{x \rightarrow +\infty} [(ax + b)e^{\frac{1}{x}} - x] = \lim_{x \rightarrow +\infty} x(ae^{\frac{1}{x}} - 1) + \lim_{x \rightarrow +\infty} be^{\frac{1}{x}} = \lim_{x \rightarrow +\infty} x(ae^{\frac{1}{x}} - 1) + b$,

因为极限存在, 所以 $\lim_{x \rightarrow +\infty} (ae^{\frac{1}{x}} - 1) = 0$, 即 $a = 1$.

而 $\lim_{x \rightarrow +\infty} x(ae^{\frac{1}{x}} - 1) + b = \lim_{x \rightarrow +\infty} \frac{e^{\frac{1}{x}} - 1}{\frac{1}{x}} + b = 1 + b = 2$, 所以 $b = 1$.

(16) 【解】 积分区域如右图阴影部分所示, 且曲线与直线的交点为

$\left(\frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{\sqrt{2}}\right)$, 则



三(16) 题图

$$\begin{aligned}\iint_D x^2 dx dy &= \int_0^{\frac{\sqrt{2}}{2}} dx \int_{\sqrt{3}x}^{\sqrt{3(1-x^2)}} x^2 dy = \sqrt{3} \int_0^{\frac{\sqrt{2}}{2}} x^2 (\sqrt{1-x^2} - x) dx \\ &= \sqrt{3} \left(\int_0^{\frac{\sqrt{2}}{2}} x^2 \sqrt{1-x^2} dx - \int_0^{\frac{\sqrt{2}}{2}} x^3 dx \right),\end{aligned}$$

$$\begin{aligned}\text{其中 } \int_0^{\frac{\sqrt{2}}{2}} x^2 \sqrt{1-x^2} dx &\stackrel{x=\sin t}{=} \int_0^{\frac{\pi}{4}} \sin^2 t \cdot \cos t d(\sin t) = \frac{1}{4} \int_0^{\frac{\pi}{4}} \sin^2 2t dt \\ &= \frac{1}{8} \int_0^{\frac{\pi}{4}} (1 - \cos 4t) dt = \frac{1}{8} \left(t - \frac{\sin 4t}{4} \right) \Big|_0^{\frac{\pi}{4}} = \frac{\pi}{32},\end{aligned}$$

$$\int_0^{\frac{\sqrt{2}}{2}} x^3 dx = \frac{x^4}{4} \Big|_0^{\frac{\sqrt{2}}{2}} = \frac{1}{16},$$

$$\text{所以 } \iint_D x^2 dx dy = \sqrt{3} \left(\frac{\pi}{32} - \frac{1}{16} \right) = \frac{\sqrt{3}(\pi - 2)}{32}.$$

(17) 【解】 设圆的周长为 x , 正三角形的周长为 y , 正方形的周长为 z , 由题设可知 $x + y + z = 2$, 则目标函数

$$S = \pi \left(\frac{x}{2\pi} \right)^2 + \frac{1}{2} \cdot \frac{\sqrt{3}}{2} \left(\frac{y}{3} \right)^2 + \left(\frac{z}{4} \right)^2 = \frac{x^2}{4\pi} + \frac{\sqrt{3}}{36} y^2 + \frac{z^2}{16}.$$

构造拉格朗日函数 $L(x, y, z; \lambda) = \frac{x^2}{4\pi} + \frac{\sqrt{3}}{36} y^2 + \frac{z^2}{16} + \lambda(x + y + z - 2)$, 对参数求导并令导函数为 0, 则

$$\begin{cases} L'_x = \frac{x}{2\pi} + \lambda = 0, \\ L'_y = \frac{2\sqrt{3}}{36} y + \lambda = 0, \\ L'_z = \frac{2z}{16} + \lambda = 0, \\ L'_\lambda = x + y + z - 2 = 0, \end{cases}$$

$$\text{解得 } x = \frac{2\pi}{\pi + 3\sqrt{3} + 4}, \quad y = \frac{6\sqrt{3}\pi}{\pi + 3\sqrt{3} + 4}, \quad z = \frac{8}{\pi + 3\sqrt{3} + 4}.$$

此时, 面积和有最小值, 即 $S = \frac{1}{\pi + 3\sqrt{3} + 4}$.

(18) 【解】 根据函数的幂级数展开, 有

$$\begin{aligned}\cos 2x &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} (2x)^{2n} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} 4^n x^{2n}, \\ -\frac{1}{(1+x)^2} &= \left(\frac{1}{1+x} \right)' = \sum_{n=0}^{\infty} [(-x)^n]' = \sum_{n=0}^{\infty} (-1)^n n x^{n-1} = \sum_{n=0}^{\infty} (-1)^{n+1} (n+1) x^n,\end{aligned}$$

所以,

$$\begin{aligned}\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} 4^n x^{2n} &= \sum_{n=0}^{\infty} a_n x^n - \sum_{n=0}^{\infty} (-1)^{n+1} (n+1) x^n \\ &= \sum_{n=0}^{\infty} [a_n - (-1)^{n+1} (n+1)] x^n.\end{aligned}$$

因为左边缺奇数项,因此 n 为奇数时, $a_n - (-1)^{n+1}(n+1) = 0$, 即 $a_n = n+1$;

$$\begin{aligned}\text{此时, } \sum_{n=0}^{\infty} [a_n - (-1)^{n+1}(n+1)] x^n &= \sum_{n=0}^{\infty} [a_{2n} - (-1)^{2n+1}(2n+1)] x^{2n} \\ &= \sum_{n=0}^{\infty} [a_{2n} + (2n+1)] x^{2n},\end{aligned}$$

$$\text{即 } a_{2n} + (2n+1) = \frac{(-1)^n}{(2n)!} 4^n, \quad a_{2n} = \frac{(-1)^n}{(2n)!} 4^n - (2n+1),$$

$$\text{综上可知 } a_n = \begin{cases} n+1, & n=1, 3, 5, \cdots, \\ \frac{(-1)^{\frac{n}{2}}}{n!} 2^n - n - 1, & n=0, 2, 4, \cdots. \end{cases}$$

(19) 【证明】 因为 $x_1 \neq 0$, 所以 $e^{x_2} = \frac{e^{x_1} - 1}{x_1}$.

由微分中值定理, 存在 $\xi \in (0, x_1)$, 使得 $\frac{e^{x_1} - 1}{x_1} = e^\xi$, 即 $e^{x_2} = e^\xi$, 因此 $0 < x_2 < x_1$.

类似, 假设 $0 < x_{n+1} < x_n$, 则

$$e^{x_{n+2}} = \frac{e^{x_{n+1}} - 1}{x_{n+1}} = e^\eta \quad (0 < \eta < x_{n+1}), \text{ 即 } 0 < x_{n+2} < x_{n+1}.$$

故 $\{x_n\}$ 是单调减少的数列, 且下有界, 从而 $\{x_n\}$ 收敛.

设 $\lim_{x \rightarrow \infty} x_n = A$, 在等式 $x_n \cdot e^{x_{n+1}} = e^{x_n} - 1$ 两边取极限, 得 $Ae^A = e^A - 1$. 显然 $A = 0$ 为其解.

又令 $f(x) = xe^x - e^x + 1$, 则 $f'(x) = xe^x$.

当 $x > 0$ 时, $f'(x) = xe^x > 0$, 函数 $f(x)$ 在 $[0, +\infty)$ 上单调增加, 所以 $A = 0$ 是方程 $Ae^A = e^A - 1$ 在 $[0, +\infty)$ 上的唯一解, 故 $\lim_{x \rightarrow \infty} x_n = 0$.

(20) 【解】 (I)

$$\text{由 } f(x_1, x_2, x_3) = 0 \text{ 得 } \begin{cases} x_1 - x_2 + x_3 = 0, \\ x_2 + x_3 = 0, \\ x_1 + ax_3 = 0, \end{cases}$$

$$\text{则系数矩阵 } \mathbf{A} = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & a \end{pmatrix} \xrightarrow{\text{初等行变换}} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & a-2 \end{pmatrix}.$$

所以当 $a \neq 2$ 时, $r(\mathbf{A}) = 3$, 方程组有唯一解 $x_1 = x_2 = x_3 = 0$.

当 $a = 2$ 时, $r(\mathbf{A}) = 2$, 方程组有无穷解, $\mathbf{X} = k \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$ (k 为任意常数).

(II) 当 $a \neq 2$ 时, 令 $\begin{cases} y_1 = x_1 - x_2 + x_3, \\ y_2 = x_2 + x_3, \\ y_3 = x_1 + ax_3, \end{cases}$ 为可逆变换, 此时规范形为 $y_1^2 + y_2^2 + y_3^2$.

当 $a = 2$ 时,

$$f(x_1, x_2, x_3) = (x_1 - x_2 + x_3)^2 + (x_2 + x_3)^2 + (x_1 + 2x_3)^2$$

$$\begin{aligned}
 &= 2x_1^2 + 2x_2^2 + 6x_3^2 - 2x_1x_2 + 6x_1x_3 \\
 &= 2\left(x_1 - \frac{x_2 - 3x_3}{2}\right)^2 + \frac{3}{2}(x_2 + x_3)^2,
 \end{aligned}$$

此时规范形为 $y_1^2 + y_2^2$.

(21) 【解】 (I) $r(A) = 2$, 而初等变换不改变矩阵的秩, 因此 B 的秩 $r(B) = 2$, 于是行列式

$$|B| = \begin{vmatrix} 1 & a & 2 \\ 0 & 1 & 1 \\ -1 & 1 & 1 \end{vmatrix} = 2 - a = 0, \text{ 即 } a = 2.$$

(II) 由 $AP = B$, 得 $P^T A^T = B^T$, 即矩阵 A^T 可经过初等行变换化为矩阵 B^T , 由分块矩阵的乘法知

$$P^T(A^T, E) = (P^T A^T, P^T) = (B^T, P^T).$$

上式表明, 对矩阵 (A^T, E) 作初等行变换化为 (B^T, P^T) , 可求得矩阵 P^T .

对下列矩阵作初等行变换, 得

$$\begin{aligned}
 (A^T : E) &= \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 2 & 3 & 7 & 0 & 1 & 0 \\ 2 & 0 & -2 & 0 & 0 & 1 \end{array} \right) \rightarrow \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 3 & -2 & 1 & 0 \\ 0 & -2 & -6 & -2 & 0 & 1 \end{array} \right) \\
 &\rightarrow \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 3 & -2 & 1 & 0 \\ 0 & 0 & 0 & -6 & 2 & 1 \end{array} \right) \rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 3 & -1 & 0 \\ 0 & 1 & 3 & -2 & 1 & 0 \\ 0 & 0 & 0 & -6 & 2 & 1 \end{array} \right) \\
 &\rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 3 & -1 & 0 \\ 2 & 1 & 1 & 4 & -1 & 0 \\ 2 & 1 & 1 & -2 & 1 & 1 \end{array} \right).
 \end{aligned}$$

$$\text{于是 } P = \begin{pmatrix} 3 & 4 & -2 \\ -1 & -1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

(22) 【解】 (I) 因为 $E(X) = 0, E(X^2) = 1, E(Y) = \lambda$, 以及 X, Y 相互独立, 故

$$\begin{aligned}
 \text{Cov}(X, Z) &= \text{Cov}(X, XY) = E(X^2 Y) - E(X)E(XY) \\
 &= E(X^2)E(Y) - E^2(X)E(Y) = \lambda.
 \end{aligned}$$

(II) 由 Y 服从参数为 λ 的泊松分布, 即 $P\{Y=j\} = \frac{\lambda^j}{j!} e^{-\lambda} (j=0, 1, 2, \dots)$, 于是 Z 的所有

可能取值为全体整数. 故 Z 的概率分布为

1) 当 k 为正整数时, 有

$$P\{Z=k\} = P\{XY=k\} = P\{X=1, Y=k\} = P\{X=1\}P\{Y=k\} = \frac{1}{2} \cdot \frac{\lambda^k}{k!} e^{-\lambda};$$

2) 当 k 为负整数时, 有

$$\begin{aligned}
 P\{Z=k\} &= P\{XY=k\} = P\{X=-1, Y=-k\} = P\{X=-1\}P\{Y=-k\} \\
 &= \frac{1}{2} \cdot \frac{\lambda^{-k}}{(-k)!} e^{-\lambda};
 \end{aligned}$$

3) 当 k 为 0 时, 有

$$\begin{aligned} P\{Z=0\} &= P\{XY=0\} = P\{X=-1, Y=0\} + P\{X=1, Y=0\} \\ &= P\{X=-1\}P\{Y=0\} + P\{X=1\}P\{Y=0\} \\ &= \frac{1}{2}e^{-\lambda} + \frac{1}{2}e^{-\lambda} = e^{-\lambda}. \end{aligned}$$

$$\text{故 } P(Z=k) = \begin{cases} \frac{1}{2} \cdot \frac{\lambda^k}{k!} e^{-\lambda}, & k=1, 2, 3, \dots, \\ e^{-\lambda}, & k=0, \\ \frac{1}{2} \cdot \frac{\lambda^{-k}}{(-k)!} e^{-\lambda}, & k=-1, -2, -3, \dots \end{cases}$$

(23) 【解】 (I) 似然函数为

$$L(x_1, x_2, \dots, x_n; \sigma) = \prod_{i=1}^n f(x_i; \sigma) = \frac{1}{2^n \sigma^n} e^{-\frac{1}{\sigma} \sum_{i=1}^n |x_i|}, \quad -\infty < x_i < +\infty, i=1, 2, \dots, n.$$

$$\text{于是 } \ln L = -n \ln 2 - n \ln \sigma - \frac{1}{\sigma} \sum_{i=1}^n |x_i|.$$

$$\text{令 } \frac{d \ln L}{d \sigma} = -\frac{n}{\sigma} + \frac{1}{\sigma^2} \sum_{i=1}^n |x_i| = 0, \text{ 得 } \sigma \text{ 的最大似然估计量为 } \hat{\sigma} = \frac{1}{n} \sum_{i=1}^n |X_i|.$$

$$\begin{aligned} \text{(II)} E(\hat{\sigma}) &= \frac{1}{n} \sum_{i=1}^n E(|X_i|) = E|X| = \int_{-\infty}^{+\infty} |x| \frac{1}{2\sigma} e^{-\frac{|x|}{\sigma}} dx = \int_0^{+\infty} \frac{x}{\sigma} e^{-\frac{x}{\sigma}} dx \\ &= - \int_0^{+\infty} x d e^{-\frac{x}{\sigma}} = -x e^{-\frac{x}{\sigma}} \Big|_0^{+\infty} + \int_0^{+\infty} e^{-\frac{x}{\sigma}} dx = \sigma. \end{aligned}$$

$$\begin{aligned} D(\hat{\sigma}) &= \frac{1}{n^2} \sum_{i=1}^n D(|X_i|) = \frac{D(|X|)}{n} = \frac{E[|X|^2 - E^2(|X|)]}{n} \\ &= \frac{1}{n} \left(\int_{-\infty}^{+\infty} |x|^2 \frac{1}{2\sigma} e^{-\frac{|x|}{\sigma}} dx - \sigma^2 \right) = \frac{1}{n} \left(\int_0^{+\infty} \frac{x^2}{\sigma} e^{-\frac{x}{\sigma}} dx - \sigma^2 \right) \\ &= \frac{1}{n} \left(\int_0^{+\infty} -x^2 d e^{-\frac{x}{\sigma}} - \sigma^2 \right) = \frac{1}{n} \left(\int_0^{+\infty} 2x e^{-\frac{x}{\sigma}} dx - \sigma^2 \right) \\ &= \frac{1}{n} (2\sigma^2 - \sigma^2) = \frac{\sigma^2}{n}. \end{aligned}$$